

1.6 [20] <§1.6> Consider two different implementations of the same instruction set architecture. The instructions can be divided into four classes according to their CPI (class A, B, C, and D). P1 with a clock rate of 2.5 GHz and CPIs of 1, 2, 3, and 3, and P2 with a clock rate of 3 GHz and CPIs of 2, 2, 2, and 2.

Given a program with a dynamic instruction count of $1.0E6$ instructions divided into classes as follows: 10% class A, 20% class B, 50% class C, and 20% class D, which implementation is faster?

- a. What is the global CPI for each implementation?
- b. Find the clock cycles required in both cases.

1.8 The Pentium 4 Prescott processor, released in 2004, had a clock rate of 3.6 GHz and voltage of 1.25 V. Assume that, on average, it consumed 10 W of static power and 90 W of dynamic power.

The Core i5 Ivy Bridge, released in 2012, had a clock rate of 3.4 GHz and voltage of 0.9 V. Assume that, on average, it consumed 30 W of static power and 40 W of dynamic power.

1.8.1 [5] <§1.7> For each processor find the average capacitive loads.

1.8.2 [5] <§1.7> Find the percentage of the total dissipated power comprised by static power and the ratio of static power to dynamic power for each technology.

1.8.3 [15] <§1.7> If the total dissipated power is to be reduced by 10%, how much should the voltage be reduced to maintain the same leakage current? Note: power is defined as the product of voltage and current.

1.11 The results of the SPEC CPU2006 bzip2 benchmark running on an AMD Barcelona has an instruction count of 2.389×10^{12} , an execution time of 750 s, and a reference time of 9650 s.

1.11.1 [5] <§1.6, 1.9> Find the CPI if the clock cycle time is 0.333 ns.

1.11.2 [5] <§1.9> Find the SPECratio.

1.11.3 [5] <§1.6, 1.9> Find the increase in CPU time if the number of instructions of the benchmark is increased by 10% without affecting the CPI.

1.11.4 [5] <§1.6, 1.9> Find the increase in CPU time if the number of instructions of the benchmark is increased by 10% and the CPI is increased by 5%.

1.11.5 [5] <§1.6, 1.9> Find the change in the SPECratio for this change.

1.11.6 [10] <§1.6> Suppose that we are developing a new version of the AMD Barcelona processor with a 4 GHz clock rate. We have added some additional instructions to the instruction set in such a way that the number of instructions has been reduced by 15%. The execution time is reduced to 700 s and the new SPECratio is 13.7. Find the new CPI.

1.11.7 [10] <§1.6> This CPI value is larger than obtained in 1.11.1 as the clock rate was increased from 3 GHz to 4 GHz. Determine whether the increase in the CPI is similar to that of the clock rate. If they are dissimilar, why?

1.11.8 [5] <§1.6> By how much has the CPU time been reduced?

1.11.9 [10] <§1.6> For a second benchmark, libquantum, assume an execution time of 960 ns, CPI of 1.61, and clock rate of 3 GHz. If the execution time is reduced by an additional 10% without affecting the CPI and with a clock rate of 4 GHz, determine the number of instructions.

1.11.10 [10] <§1.6> Determine the clock rate required to give a further 10% reduction in CPU time while maintaining the number of instructions and with the CPI unchanged.

1.11.11 [10] <§1.6> Determine the clock rate if the CPI is reduced by 15% and the CPU time by 20% while the number of instructions is unchanged.

1.14 Assume a program requires the execution of 50×10^6 FP instructions, 110×10^6 INT instructions, 80×10^6 L/S instructions, and 16×10^6 branch instructions. The CPI for each type of instruction is 1, 1, 4, and 2, respectively. Assume that the processor has a 2 GHz clock rate.

1.14.1 [10] <\$1.10> By how much must we improve the CPI of FP instructions if we want the program to run two times faster?

1.14.2 [10] <\$1.10> By how much must we improve the CPI of L/S instructions if we want the program to run two times faster?

1.14.3 [5] <\$1.10> By how much is the execution time of the program improved if the CPI of INT and FP instructions is reduced by 40% and the CPI of L/S and Branch is reduced by 30%?

2.4 [5] <§§2.2, 2.3> For the MIPS assembly instructions below, what is the corresponding C statement? Assume that the variables *f*, *g*, *h*, *i*, and *j* are assigned to registers *\$s0*, *\$s1*, *\$s2*, *\$s3*, and *\$s4*, respectively. Assume that the base address of the arrays *A* and *B* are in registers *\$s6* and *\$s7*, respectively.

```
sll    $t0, $s0, 2      # $t0 = f * 4
add    $t0, $s6, $t0    # $t0 = &A[f]
sll    $t1, $s1, 2      # $t1 = g * 4
add    $t1, $s7, $t1    # $t1 = &B[g]
lw     $s0, 0($t0)      # f = A[f]
addi   $t2, $t0, 4
lw     $t0, 0($t2)
add    $t0, $t0, $s0
sw     $t0, 0($t1)
```

2.9 [5] <§§2.2, 2.3> Translate the following C code to MIPS. Assume that the variables *f*, *g*, *h*, *i*, and *j* are assigned to registers *\$s0*, *\$s1*, *\$s2*, *\$s3*, and *\$s4*, respectively. Assume that the base address of the arrays *A* and *B* are in registers *\$s6* and *\$s7*, respectively. Assume that the elements of the arrays *A* and *B* are 4-byte words:

```
B[8] = A[i] + A[j];
```